Sigmatronics Innovation

Sigmatronics Innovation Pvt Ltd provides industrial IoT solutions and products. Our products help industries & organizations get insights from their assets in real-time so they can act proactively.

OUR OFFERINGS

- SIGMATRONICS
- Research & Development
- Electronics Engineering
- IoT Products & Solutions
- Industrial Design
- Rapid Prototyping

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SEGMENTS WE SERVE

Our products have found a wide use in residential, commercial & industrial segments. Along with them our services have opened limitless possibilities for our clients on various business verticals.

INDUSTRIAL

COMMERCIAL

RESIDENTIAL

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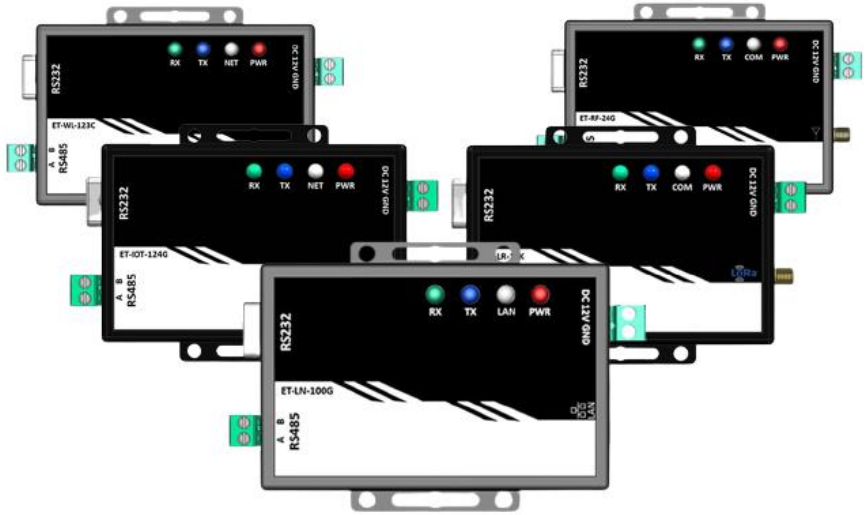
IoT GATEWAYS



Smart IoT Data Loggers

ET Series IoT data loggers are our smart plug 'n' play gateways which can be connected with more than 100 brands/models of solar PV inverters, energy meters, generators, weather monitoring stations and MODBUS enabled sensors.

We deliver robust, scalable & flexible way for data acquisition from your assets, hence allowing you to execute installation & operations with minimum hassles.

AVAILABLE IN

4G-LTE | GPRS | LoRa | RF | Wi-Fi | ETHERNET

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GATEWAY SPECIFICATIONS

	GPRS	4G LTE	RF	LoRa	Wi-Fi	Ethernet
COM PORT	RS232* RS485	RS232* RS485	RS232* RS485	RS232* RS485	RS232* RS485	RS232* RS485
SLAVES	8	8	8	8	31	31
PARAMETERS	< 60	< 60	< 60	< 60	< 300	< 300
RANGE	GSM	GSM	1KM	10KM	INTRANET	INTRANET
PROTOCOL SUPPORT	MODBUS	MODBUS	MODBUS	MODBUS	MODBUS	MODBUS
INPUT VOLTAGE	12V DC/ 230V AC*	12V DC/ 230V AC*	12V DC/ 230V AC*	12V DC/ 230V AC*	12V DC/ 230V AC*	12V DC/ 230V AC*
NETWORK TOPOLOGY	P2P	P2P	TREE	STAR	STAR	BUS/STAR
IP RATING	IP55/ IP67*	IP55/ IP67*	IP55/ IP67*	IP55/ IP67*	IP55/ IP67*	IP55/ IP67*

* Options are available on request

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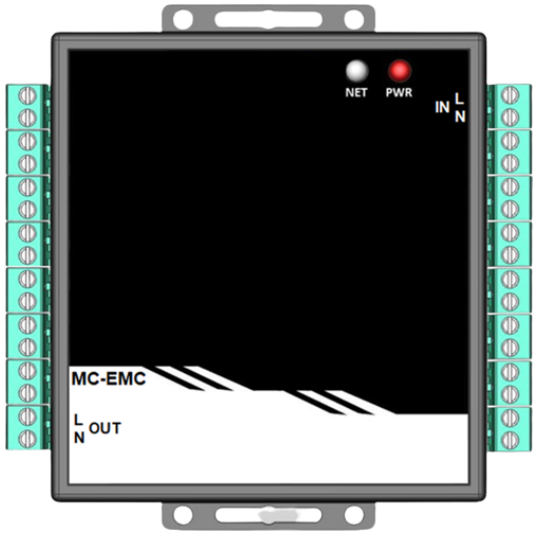
ENERGY MANAGEMENT CONTROLLER



Energy Management Controller

Energy management controllers are used to control & Turn ON/OFF connected appliances over Wi-Fi/Internet. It also monitors the realtime power consumption of each appliance and sends the data to remote server.

Benefits:

- Track power consumption from all the appliances
- Identify any faulty appliances
- schedule when devices turns on/off
- interactive dashboards.

Our EMC devices have small form factor & gets connected to appliances directly inside electrical distribution boxes. Configurable site as per user settings for faster setup without any hassle.

FEATURES

- Connect over Wi-Fi
- Multiple I/O Channels
- Integrated Power Sensors
- Switching & Monitoring Dashboard
- Schedule Appliance Switching

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SIGMATRONICS : IoT Analytics Platform

Sigmatronics
The smartest way forward

A SaaS based IoT Analytics platform for monitoring, managing & control of your assets. Now track performance, spot or predict failures and provide proactive maintenance. Get full control over your system remotely.

Connect your assets from solar plants, utility meters, weather monitors, appliances, sensors etc. & manage all at one place

- Energy Meters
- Solar PV Systems
- Generators
- Weather Stations
- Appliances
- Sensors

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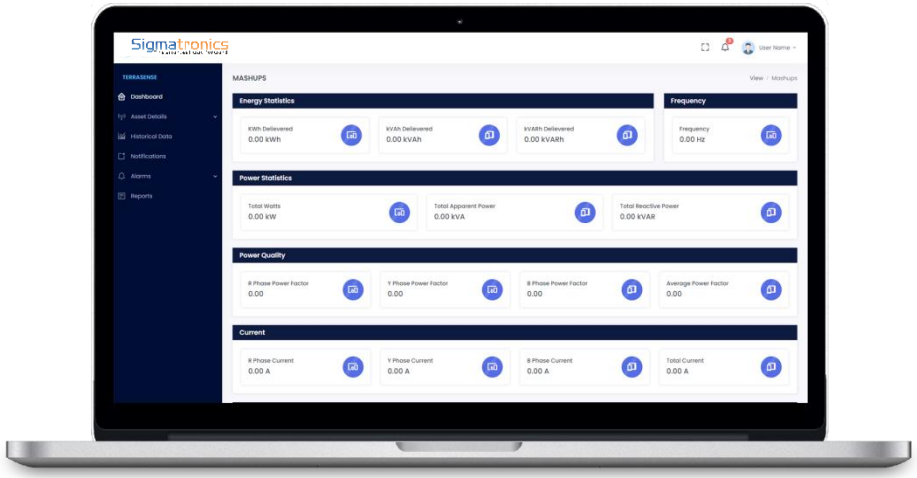
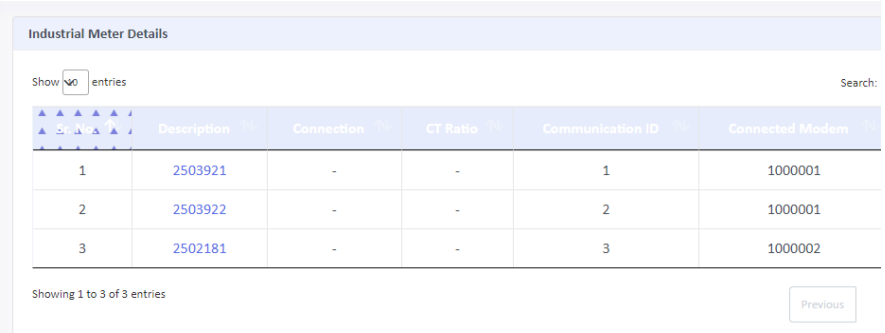
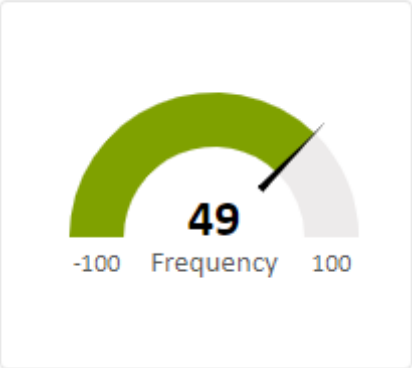
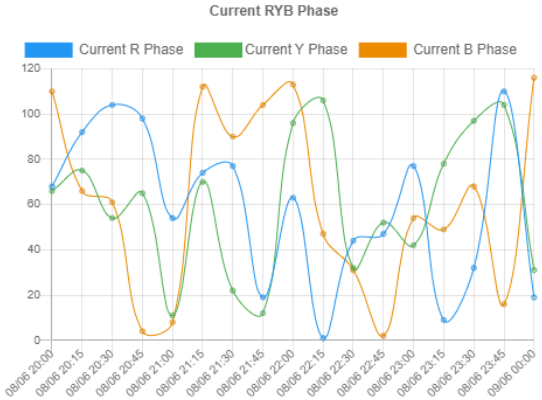
VISUALIZATIONS









Visualize all the data from your assets on dedicated dashboards with tables, charts, graphs & gauges.

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SIGMATRONICS FEATURES

Connect | Visualize | Manage

Mashups
Visualize all your important data from various assets in one place with intuitive screen.

Trends
Enabling historical data analysis to analyze plant as well as individual asset performance on daily, monthly & yearly basis.

Realtime Data
Get all the data from connected sensors & devices on dedicated real-time dashboards.

Alarm Engine
Configure simple and complex SMS & email alerts to drive proactive response towards abnormality in asset data.

Insightful Reports
Generate and schedule smart analytical and customized reports in PDF & Spreadsheet.

Billings
Configurable tariffs, Bill Generation, Recharge for utilities on the go with integrated payment gateways.

Manage Assets
Manage all the connected sensors, assets, controllers, data integrations from a single interface.

Notifications
Get real time instant fault alerts via SMS & email in case of system failure & all other issues occurring in the assets.

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SAAS SUBSCRIPTION PLANS

ADVANCED

- ✓ Parameters Limit: < 30
- ✓ Data Receiving Frequency: 10 min+
- ✓ Real-time Dashboard: Available
- ✓ Custom Dashboard: Available
- ✓ Notification & Alerts: Available
- ✓ Payment Gateway: On Request
- ✓ Data Retention: 365 day

ENTERPRISE

- ✓ Parameters Limit: Custom
- ✓ Data Receiving Frequency: Custom
- ✓ Real-time Dashboard: Available
- ✓ Custom Dashboard: Available
- ✓ Notification & Alerts: Available
- ✓ Monthly Email/SMS Limit : Custom
- ✓ Payment Gateway: On Request
- ✓ Alarm Engine: Custom
- ✓ Condition Based Alerts: Available
- ✓ Data Retention: Custom

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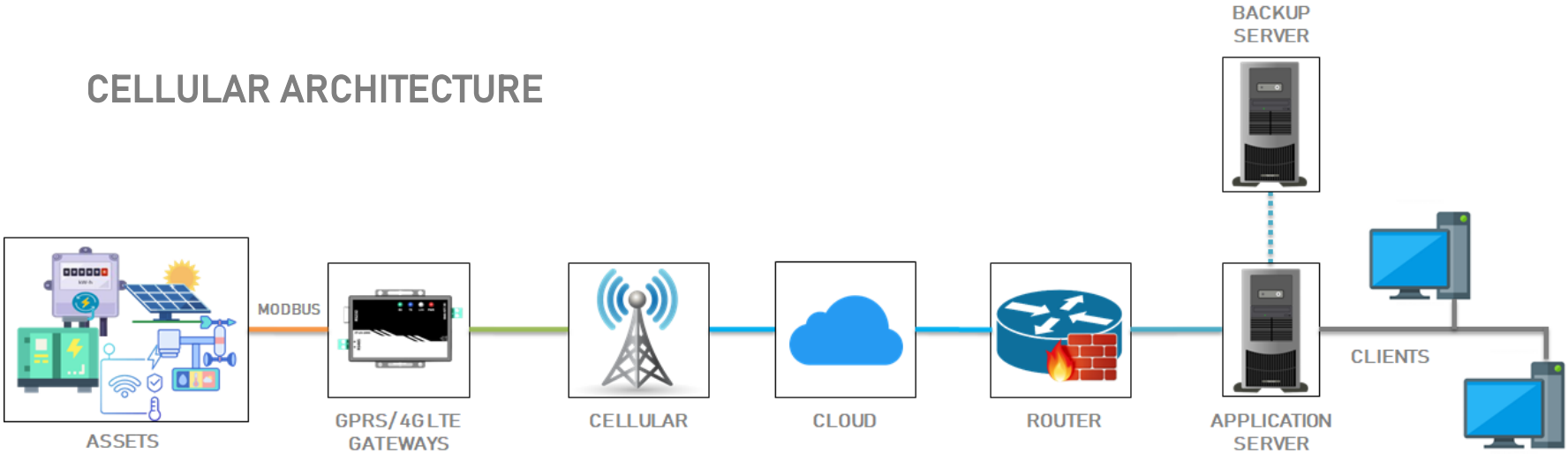
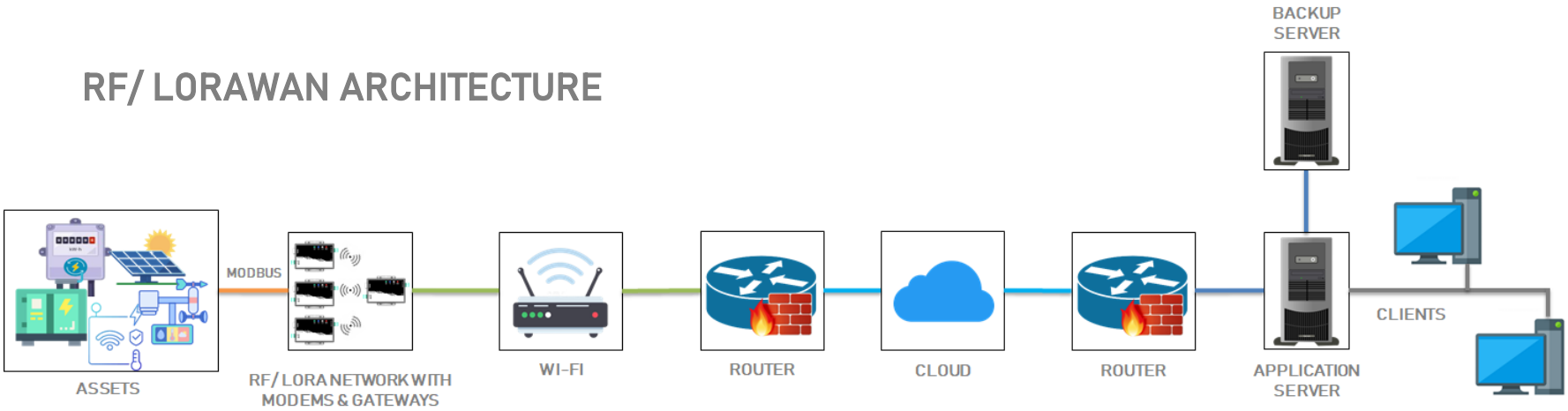
ARCHITECTURES

CELLULAR ARCHITECTURE



RF/ LORAWAN ARCHITECTURE



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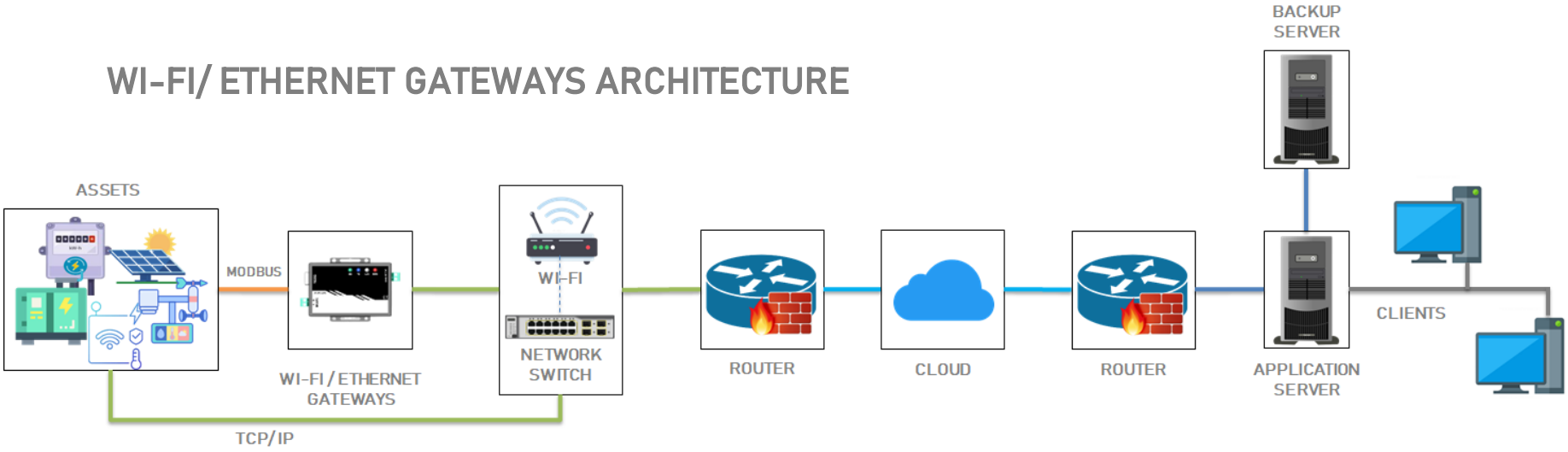
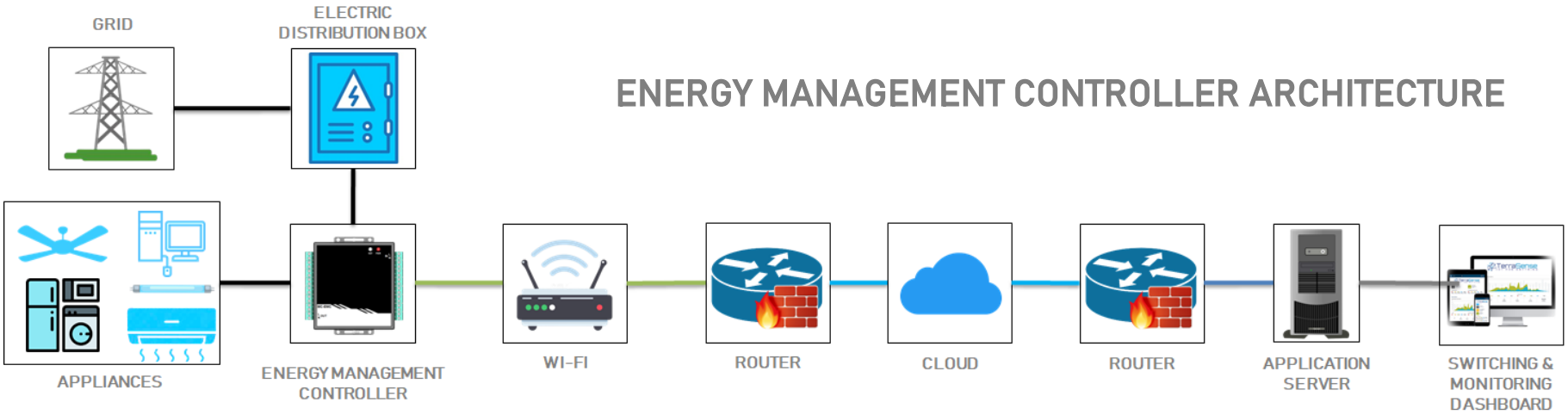
ARCHITECTURES

WI-FI/ ETHERNET GATEWAYS ARCHITECTURE



ENERGY MANAGEMENT CONTROLLER ARCHITECTURE



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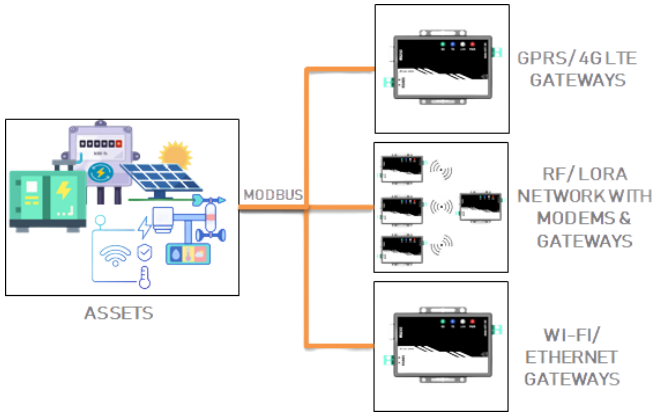
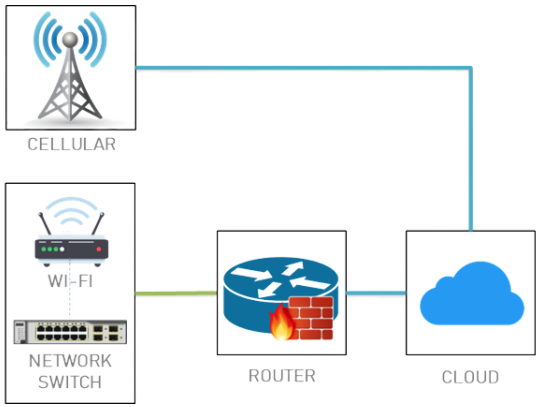
HOW IT WORKS

1



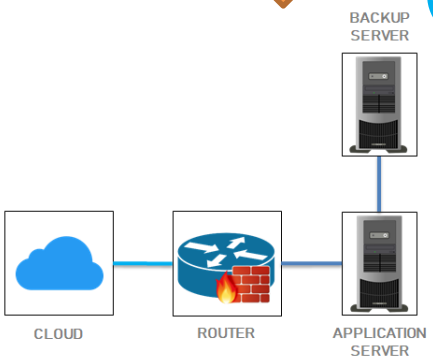
Connect your assets supporting modbus with required communication gateways based on the site. Configure basic settings like slave id, machine type or group and save settings to device.

2



Gateways connect to the internet over cellular data connectivity or with network switchs/ wi-fi networks. Lan networks can be set-up if it is not available or existing network can not be used.

3



Application server collects the data received from gateways at predefined intervals & creates backup at eod/scheduled time.

4

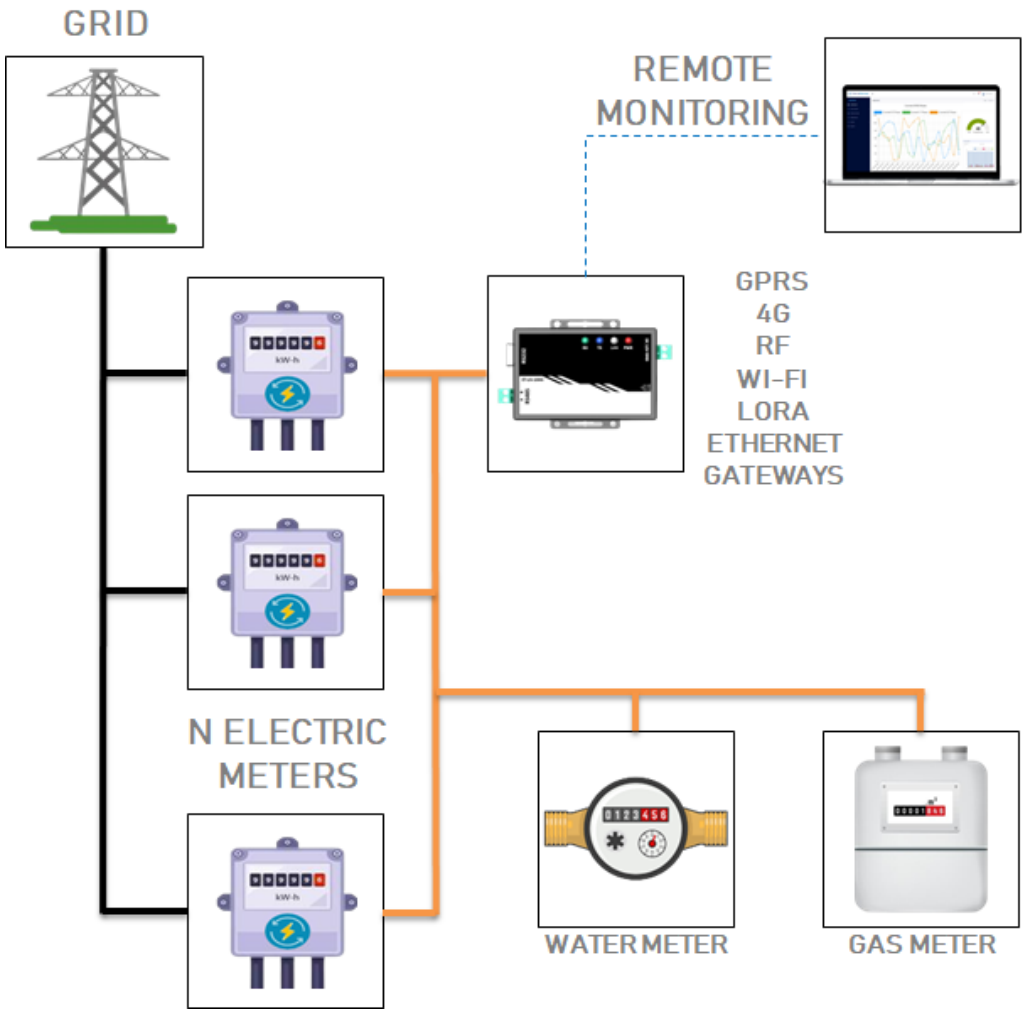
Users can setup alarms to monitor specific datasets & save the settings. Our smart analytics engine works upon the data to create insights.

5

Data is shown graphically on client side over the internet or locally on premise. Users can visualize data generate reports, bills, spot failures & track performance in realtime.

Automatic Meter Reading

SIGMATRONICS is a high potential platform to remotely monitor energy consumption from individual meters in real-time. Our IoT gateways connect with various brands/models of utility meters to collect realtime data and sends it across to our platform for smart analytics. Selection of gateways as per users site requirement.



Benefits

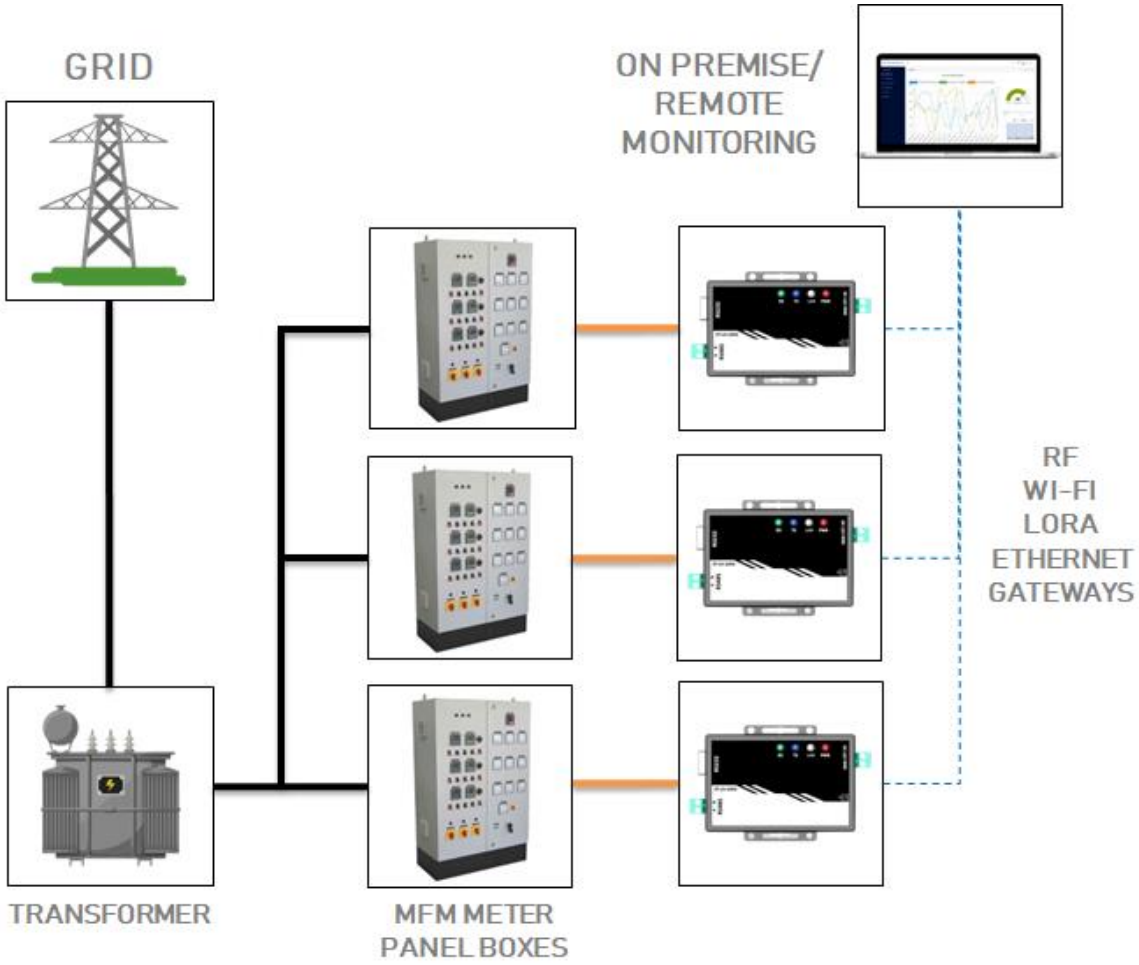
- ✓ Track data in realtime from multiple meters (electricity/ water/ gas)
- ✓ Spot & predict failures instantly
- ✓ Graphical Illustration
- ✓ Automated Reports
- ✓ Alarm & Notifications
- ✓ Prepaid & postpaid metering
- ✓ Tariff module for cost calculation

Working

The AMR system has IoT gateways that polls connected energy meters to collect electrical data & sends to web server on premise or on cloud.

Industrial Energy Monitoring

Every plant or facility wastes energy in hundreds of places and processes. We help you identify this wastage and reduce it. SIGMATRONICS helps you reduce financial and environmental impact of wasteful energy consumption by effectively managing energy use and the factors that affect it. We help both small and large organizations in measuring, analyzing and reducing energy costs by up to 20%.



Benefits

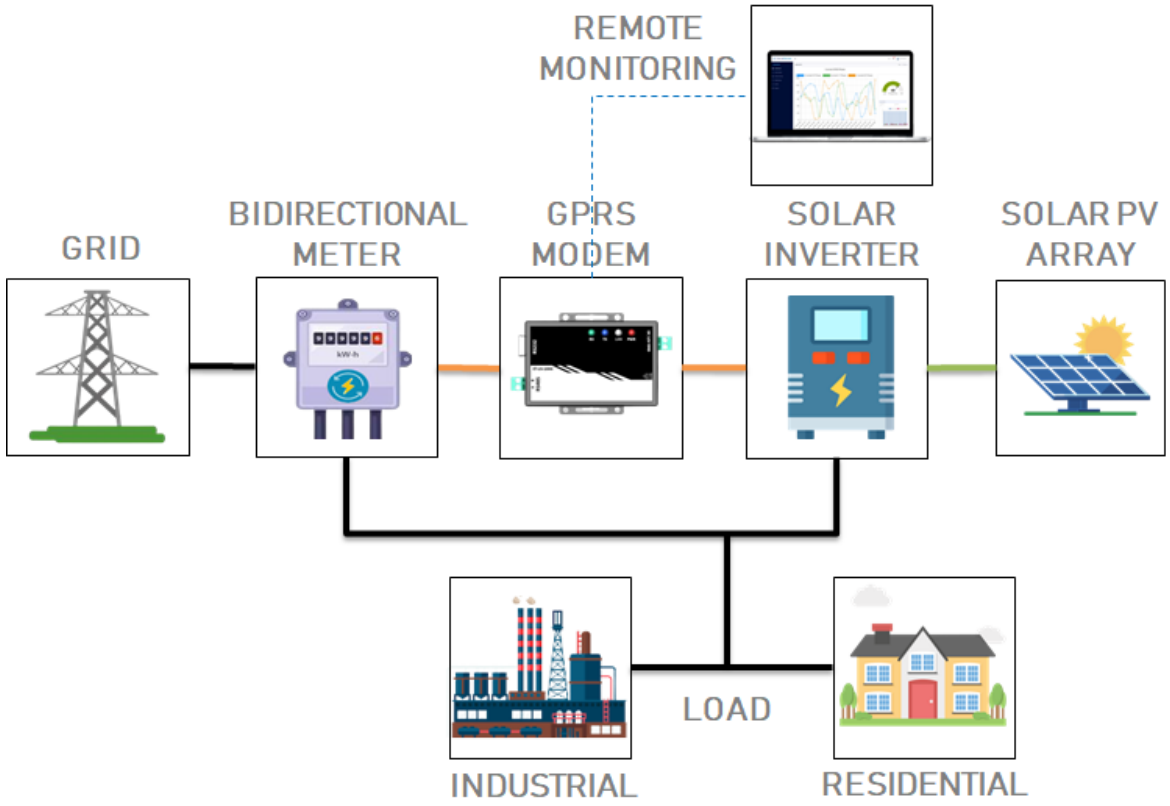
- ✓ Track electrical data in realtime from multiple energy meter
- ✓ Spot & predict failures instantly
- ✓ Graphical Illustration
- ✓ Automated Reports
- ✓ Alarm & Notifications
- ✓ Mimic Diagrams
- ✓ Plant/floor wise data visualization

Working

Multiple meters from panel boxes are polled by dedicated panel gateways & collected data is sent to on premise or remote web server.

Solar Energy Monitoring

Real-time solar power monitoring for Rooftop and Ground Mounted solar power plants with SIGMATRONICS platform to ensure the performance of the installed solar plant. Our remote IoT based interactive platform, visualizations & alarms engine combined with advanced analytics helps maximizing the solar yield.



Monitoring

- ✓ Grid Energy Parameters
- ✓ Solar Inverter Parameters

Analytics

- ✓ Import/Export/Generation
- ✓ Actual Consumption in kWh
- ✓ Performance Ratio of solar

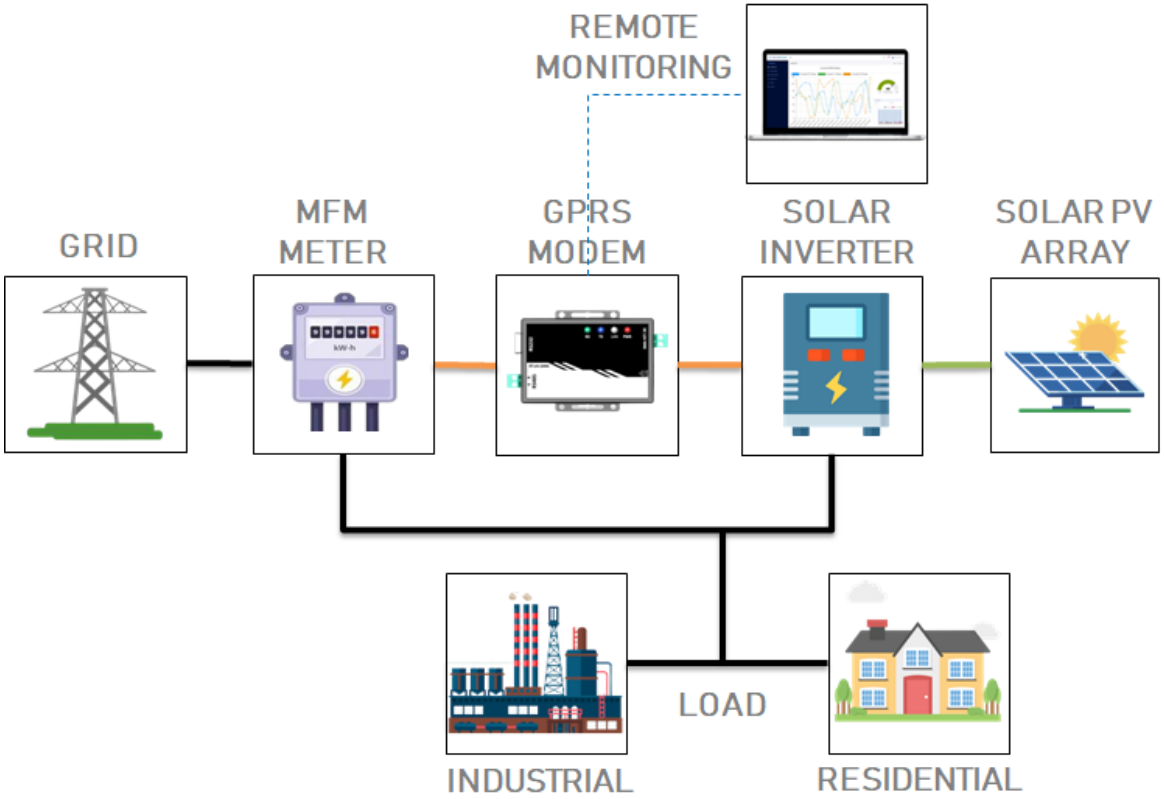
Benefits

- ✓ Track performance in realtime from multiple inverters & meters
- ✓ Spot & predict failures instantly
- ✓ Provide proactive maintenance
- ✓ Increase generation efficiency
- ✓ Full control of system

ZERO EXPORT CONTROL

Solar Monitoring With Zero Export Control

In case of unavailability of net metering, to use solar as per load requirement, we need to sync solar with the grid. In this solution, solar will take only reference voltage from the grid, but does not export power to the grid. This way the user would be able to use grid-connected solar plant without export



Benefits

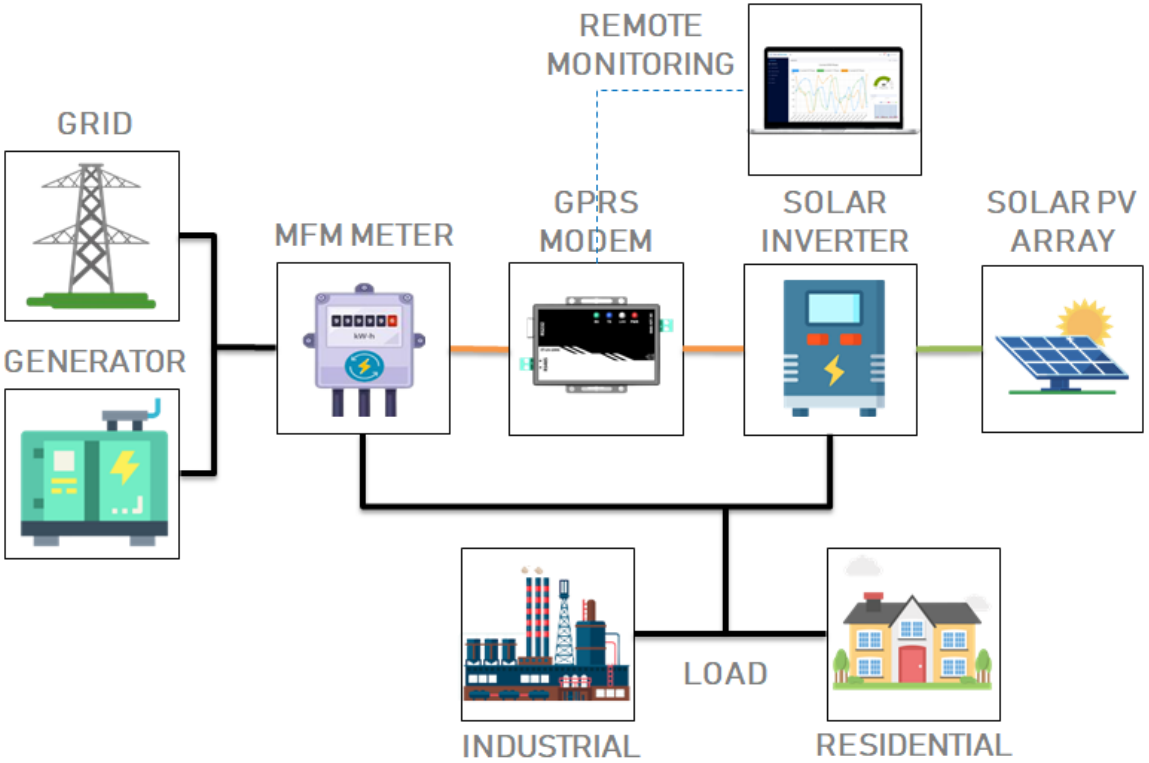
- ✓ Smart Energy Control
- ✓ Reverse Feed Protection
- ✓ Graphical Illustration
- ✓ Remote Monitoring
- ✓ Deemed Generation

Working

The zero export solution polls connected energy meters and inverters to determine the current load and help user to use solar as per demanded load. This solution will throttle the power of solar inverters as per site load and there will be no export to grid.

DG-PV Synchronization

Our IoT gateway smartly distributes the total demanded load between Solar and DG in such a way that the DG runs at the minimum possible load, securing against reverse feeding and protecting the inverters from over-frequency damage, while also increasing fuel efficiency.



Benefits

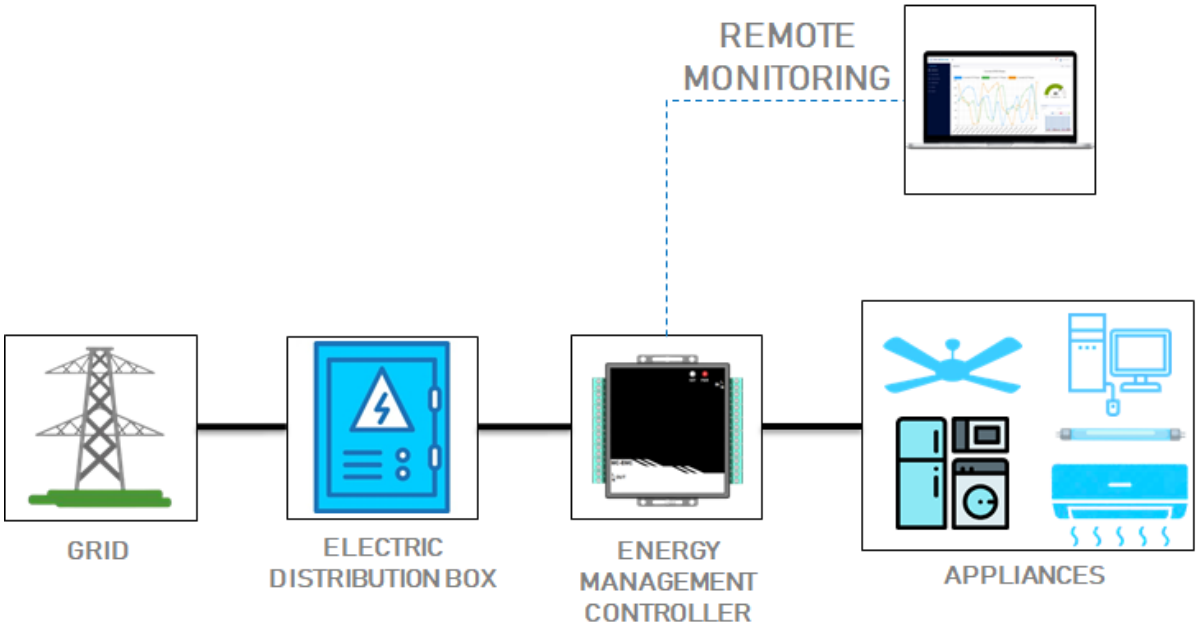
- ✓ Smart Energy Control
- ✓ Reverse Feed Protection
- ✓ Graphical Illustration
- ✓ Remote Monitoring
- ✓ Deemed Generation

Working

Energy meter is installed at DG panel. When grid power is unavailable & we get ON status from DG, the DG PV controller throttles the power of solar Inverter as per site load. This will result in the minimum fuel consumption of DG which runs at 30-35% capacity & avoid reverse power protection with solar generation optimization.

Energy Control & Management

Energy Management Controller installs in any electrical panel. It supports both single-phase and three-phase. The EMC monitors the total energy usage & controls switching of each device using web dashboard. Users can choose multichannel 8/16 options for their needs.



Benefits

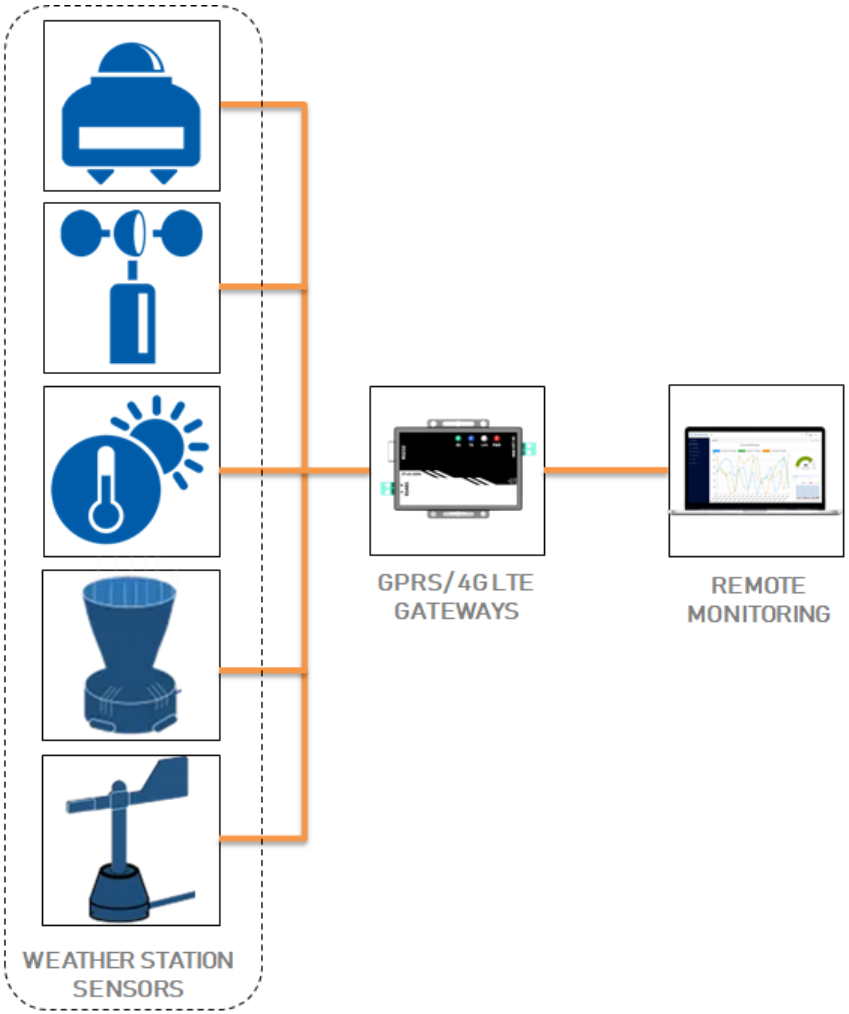
- ✓ Web based switching controls
- ✓ Use with any appliance
- ✓ Remotely monitors energy use
- ✓ Conserve energy, saves money
- ✓ Reduces energy waste
- ✓ Switching scheduler for smart control of appliances.

Working

EMC is installed inside electric panel (By a professional electrician) & device configured with Wi-Fi settings. Device starts sending data to cloud server & all the devices are visible on dashboard. User can control, schedule appliances using the dashboard & conserve energy.

Weather Station Monitoring Systems

Integrate with any Weather Monitoring System with multiple sensors like Irradiation (Pyranometer), Module/Ambient Temperature, Humidity, Wind Speed & Direction, Rainfall etc. sensors that support Modbus. Our IoT gateways with cellular connectivity pull data from all.



Benefits

- ✓ Tracks real-time and historical weather conditions of the system and helps enable a comprehensive evaluation of system performance
- ✓ Captures weather parameters like solar radiation, wind speed & Direction, Humidity, air and module temperature, Air Pressure etc.
- ✓ Visualization of Solar Irradiation v/s Power v/s Temperatures Curves.
- ✓ Calculate PR, Expected vs. Actual Power Output, Module Efficiency, overall system efficiency etc.
- ✓ Immediate identification of any anomalies in the system.

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ROI/ IMPACT

ENERGY SAVINGS	~20%
FAULT TRACKING	~40%
SYSTEM PERFORMANCE	~30%
ENERGY EFFICIENCY	~30%
ENERGY CONSUMPTION	~30%
ENERGY WASTAGE	~30%
CARBON FOOTPRINT	~25%
MAINTENACE COSTS	~20%

ROI PAYBACK TIME
12-18 MONTHS

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TOWNSHIP AMR SYSTEM

Township Automated Meter Reading Solution for Big Power Generation Company

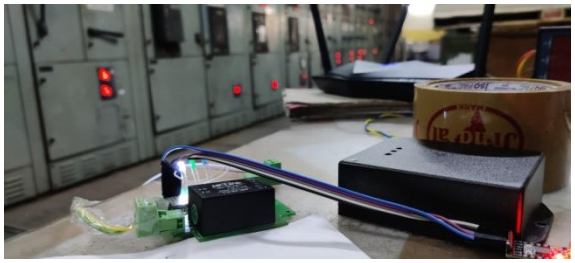
A big township of central India's big power plant wanted to know energy consumption by their residents & commercial shop owners on a daily basis. We implemented AMR on 650+ meters distributed in 50 buildings. For remote data transfer complete township was covered with Wi-Fi network and our IoT gateways started collecting electrical data from all the meters & sent to on premise server.

Key Outcomes

- ✓ Data from all the meters were polled at 15 min frequency and sent to on premise server.
- ✓ The township's electrical department is now able to track energy consumption by each of the residents & shop owners in the township.
- ✓ Billing module helps the electrical department to provide electricity bills to visitors/ commercial shop owners.

Industrial Energy Monitoring Solution for Polyfilms Manufacturing Industry

A big industry from western India's big polyfilms manufacturing industry wanted to know electrical data from various assets on a daily basis. We implemented AMR on 50+ meters distributed in 3 electrical panel rooms. For remote data transfer all panel rooms were covered with Wi-Fi network and our IoT gateways started collecting electrical data from all the meters.



Benefits

- ✓ Data from all the meters were polled at 15 min frequency and sent to on premise server.
- ✓ Electrical department now tracks real-time electrical data and conditions of all the assets Our on premise analytics server helps them with a comprehensive evaluation of assets performance
- ✓ Data visualization & report generation module helped the industry eliminate manual data recording process and thereby eliminating human errors in important data records.

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